

USAF ACADEMY AF, CO, USA

WMO: 745310

Lat: 38.967N

Lon: 104.817W

Elev: 2003

StdP: 79.46

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 93065

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.6	-15.7	-24.8	0.5	-5.0	-22.4	0.6	-4.2	16.0	8.8	12.9	1.5	3.5	0	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	31.8	13.5	30.4	13.6	29.1	13.5	16.9	24.1	16.2	23.7	15.6	23.2	5.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.8	17.9	14.2	13.0	17.5	13.4	12.3	17.7	55.1	24.2	52.9	23.2	50.8	23.4	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.0	-24.3	34.2	3.2	1.1	-26.6	35.0	-28.4	35.6	-30.2	36.2	-32.5	37.0		
(5)			-24.6	18.7	3.1	0.6	-26.9	19.1	-28.7	19.5	-30.4	19.8	-32.7	20.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-1.2	-0.9	3.9	6.8	11.3	18.0	20.4	19.2	15.9	8.3	3.2	-2.2	(6)
(7)		DBStd	9.29	5.71	6.34	5.42	4.77	4.65	3.37	2.27	2.36	3.54	5.34	5.48	6.09	(7)
(8)		HDD10.0	1694	347	307	198	117	43	1	0	0	4	91	208	379	(8)
(9)		HDD18.3	3717	606	539	447	347	221	44	8	18	87	311	454	637	(9)
(10)		CDD10.0	1187	1	1	9	21	82	241	324	286	179	39	4	1	(10)
(11)		CDD18.3	167	0	0	0	0	1	35	74	45	12	0	0	0	(11)
(12)		CDH23.3	3103	0	0	1	4	101	807	1078	718	371	24	0	0	(12)
(13)		CDH26.7	991	0	0	0	0	14	298	402	202	73	2	0	0	(13)
(14)	Wind	WSAvg	4.1	3.9	4.5	4.5	4.9	4.4	4.3	3.7	3.5	3.8	4.0	4.0	4.0	(14)
(15)	Precipitation	PrecAvg	522	14	15	34	63	64	51	90	86	41	27	17	14	(15)
(16)		PrecMax	765	42	43	93	212	252	151	165	150	148	68	74	50	(16)
(17)		PrecMin	280	3	3	8	6	16	4	30	14	1	2	3	6	(17)
(18)		PrecStd	111	8	9	21	44	50	35	37	32	31	18	15	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.5	22.3	24.0	28.4	34.2	33.2	31.9	30.6	25.8	21.1	16.4	(19)
(20)			MCWB	4.8	5.6	7.0	7.9	11.1	12.5	14.1	13.7	13.3	9.6	6.7	4.6	(20)
(21)		2%	DB	14.0	15.1	19.8	21.8	26.2	32.1	31.8	30.3	28.6	23.6	18.7	13.5	(21)
(22)			MCWB	3.5	3.9	5.9	7.6	9.8	12.4	14.0	14.0	12.2	9.0	5.9	3.2	(22)
(23)		5%	DB	11.7	13.0	17.5	20.0	24.0	30.1	30.6	29.0	27.2	21.8	16.4	11.3	(23)
(24)			MCWB	2.4	3.1	5.0	6.9	9.7	12.2	14.3	14.0	12.1	8.5	5.2	2.3	(24)
(25)		10%	DB	9.1	10.4	15.0	17.7	21.6	28.1	29.2	27.5	25.7	19.5	14.0	8.9	(25)
(26)			MCWB	1.2	1.7	4.1	6.2	9.5	12.2	14.5	14.1	11.9	8.0	4.0	1.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	5.9	7.5	10.2	13.4	16.8	18.0	17.8	16.0	12.7	7.5	5.0	(27)
(28)			MCDWB	15.6	17.4	20.6	16.3	21.4	24.6	24.0	24.4	23.9	17.6	18.9	14.9	(28)
(29)		2%	WB	3.8	4.4	6.5	8.9	12.2	15.6	17.1	16.8	14.7	11.1	6.5	3.6	(29)
(30)			MCDWB	12.9	14.1	18.4	17.3	20.3	23.6	24.4	23.7	22.2	19.0	17.7	12.7	(30)
(31)		5%	WB	2.6	3.2	5.4	7.9	11.3	14.7	16.5	16.0	13.9	9.7	5.5	2.5	(31)
(32)			MCDWB	11.2	12.6	16.1	17.2	19.2	23.3	23.9	23.3	21.6	18.0	15.7	10.8	(32)
(33)		10%	WB	1.3	1.9	4.4	6.9	10.4	14.0	15.9	15.4	13.1	8.5	4.3	1.2	(33)
(34)			MCDWB	8.9	10.2	14.0	16.4	18.4	23.0	23.4	23.0	21.4	18.3	13.3	8.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.2	14.2	15.5	14.7	14.4	16.9	15.6	15.3	15.8	15.8	16.0	14.6	(35)
(36)			MCDBR	17.7	17.5	18.8	19.1	19.4	20.5	18.6	18.0	18.6	20.0	19.5	18.3	(36)
(37)			MCWBR	9.5	9.0	8.6	8.2	7.2	6.8	5.7	5.7	6.7	8.6	9.3	10.2	(37)
(38)		5% WB	MCDWB	17.3	17.1	18.1	17.2	16.1	16.3	14.6	14.4	15.5	16.8	18.6	17.7	(38)
(39)			MCWBR	9.6	8.9	8.7	7.9	6.7	6.5	5.4	5.7	6.3	7.6	9.2	10.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.230	0.234	0.255	0.278	0.298	0.310	0.337	0.326	0.292	0.260	0.239	0.231	(40)	
(41)		taud	2.635	2.652	2.595	2.542	2.548	2.532	2.501	2.520	2.592	2.661	2.642	2.629	(41)	
(42)		Ebn at Noon	985	1021	1018	1002	979	963	933	939	961	973	964	959	(42)	
(43)		Edh at Noon	66	75	88	99	101	103	105	101	88	73	65	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.72	3.69	4.91	5.88	6.50	7.26	6.83	6.20	5.39	4.05	2.95	2.39	(44)	
(45)		RadStd	0.13	0.22	0.36	0.38	0.49	0.34	0.29	0.22	0.28	0.24	0.16	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+69	+21

Nomenclature: See separate page